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ABOUT DIABETES

Genetics of Diabetes

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Learn the Genetics of Diabetes

If you're living with diabetes, you probably have a lot of questions. You've probably wondered how you developed diabetes. You may worry that your children will develop it, too. You're not the only one asking these questions—and we are here to help.

What leads to diabetes?

Type 1 and type 2 diabetes have different causes, but there are two factors that are important in both. You **inherit** a predisposition to the disease, then something in your **environment** triggers it.

That's right: genes alone are not enough. One proof of this is identical twins. Identical twins have identical genes. Yet when one twin has type 1 diabetes, the other gets the disease, at most, only *half* the time. When one twin has type 2 diabetes, the other's risk is three in four at most.

Is type 1 diabetes genetic?

In most cases of type 1 diabetes, people need to inherit risk factors from both parents. We think these factors must be more common in white people because white people have the highest rate of type 1 diabetes.

Because most people who are at risk do not get diabetes, researchers want to find out what the environmental triggers are. One trigger might be related to cold weather. Type 1 diabetes develops more often in winter than summer and is more common in places with cold climates. Another trigger might be viruses. It's possible that a virus that has only mild effects on most people triggers type 1 diabetes in others. Early diet may also play a role. For example, type 1 diabetes is less common in people who were breastfed and in those who first ate solid foods at later ages.

In many people, the development of type 1 diabetes seems to take many years. In experiments that follow relatives of people with type 1 diabetes, researchers have found that most of those who later got diabetes had

certain autoantibodies, or proteins that destroy bacteria or viruses (antibodies "gone bad" that attack the body's own tissues), in their blood for years before they are diagnosed.

Your Child's Risk

If you are a man with type 1 diabetes, the odds of your child developing diabetes are **1 in 17**. If you are a woman with type 1 diabetes and your child was born before you were 25, your child's risk is **1 in 25**; if your child was born after you turned 25, your child's risk is **1 in 100**.

Your child's risk is doubled if you developed diabetes before age 11. If both you and your partner have type 1 diabetes, the risk is between **1 in 10 and 1 in 4**.

There is an exception to these numbers: about one in every seven people with type 1 diabetes has a condition called type 2 polyglandular autoimmune syndrome. In addition to having diabetes, these people also have thyroid disease and a poorly working adrenal gland—some also have other immune system disorders. If you have this syndrome, your child's risk of getting the syndrome and developing type 1 diabetes, is **one in two**.

Researchers are learning how to predict a person's odds of getting diabetes. For example, most white people with type 1 diabetes have genes called HLA-DR3 or HLA-DR4, which are linked to autoimmune disease. If you and your child are white and share these genes, your child's risk is higher. Suspect genes in other ethnic groups are less well-studied, however, scientists believe the HLA-DR7 gene may put African Americans at risk, and the HLA-DR9 gene may put Japanese people at risk.

An antibodies test can be done for children who have siblings with type 1 diabetes. This test measures antibodies to insulin, to islet cells in the pancreas or to an enzyme called glutamic acid decarboxylase (GAD).

High levels can indicate that a child has a higher risk of developing type 1 diabetes.

If you think your child might have type 1 diabetes, **contact your doctor**.

If a member of your family has type 1 diabetes, you may be eligible for a risk screening offered through the [TrialNet Pathway to Prevention Study](#) . TrialNet risk screening is free to relatives of people with type 1, and uses a simple blood test that can detect your risk of type 1 diabetes years before symptoms appear. If you are in the early stages of type 1 diabetes, you may also be eligible for a prevention study. [Learn more about how to get screened.](#) 

Is type 2 diabetes genetic?

Type 2 diabetes has a stronger link to family history and lineage than type 1, and studies of twins have shown that genetics play a very strong role in the development of type 2 diabetes. Race can also play a role.

Yet it also depends on environmental factors. Lifestyle also influences the development of type 2 diabetes. Obesity tends to run in families, and families often have similar eating and exercise habits.

If you have a family history of type 2 diabetes, it may be difficult to figure out whether your diabetes is due to lifestyle factors or genetics. Most likely it is due to both. However, don't lose heart! Studies show that it is possible to delay or prevent type 2 diabetes by exercising and losing weight. [Learn how you can prevent or delay type 2 diabetes.](#)

Your Child's Risk

Type 2 diabetes runs in families. In part, this is due to children learning bad habits—eating a poor diet, not exercising—from their parents. But there is also a genetic basis. The good news is, like in adults, it is possible to delay or prevent type 2 diabetes in youth by encouraging healthy food choices, exercise and weight loss.

More Information on Genetics

If you would like to learn more about the genetics of all forms of diabetes, the National Institutes of Health has published [*The Genetic Landscape of Diabetes*](#)[↗]. This free online book provides an overview of the current knowledge about the genetics of type 1 and type 2 diabetes, as well other less common forms of diabetes. The book is written for health care professionals and for people with diabetes interested in learning more about the disease.

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